29/07/2026

CODE-B
Phase-1

Aakash
Medical | IIT-JEE | Foundations

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TERM EXAM FOR NEET-2027 (XII STUDYING) PH-I

MM: 720 **TE-02B** **Time: 180 Mins.**

Answers

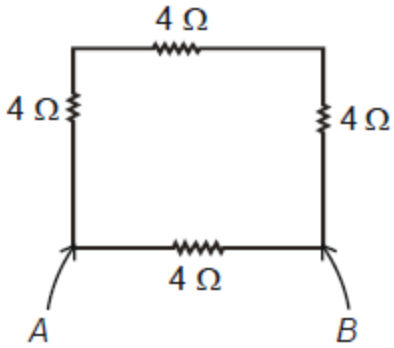
1. (2)	37. (1)	73. (2)	109. (3)	145. (2)
2. (2)	38. (2)	74. (1)	110. (2)	146. (2)
3. (1)	39. (4)	75. (1)	111. (3)	147. (3)
4. (2)	40. (1)	76. (1)	112. (1)	148. (1)
5. (4)	41. (3)	77. (4)	113. (2)	149. (3)
6. (3)	42. (3)	78. (1)	114. (3)	150. (1)
7. (1)	43. (2)	79. (2)	115. (4)	151. (4)
8. (1)	44. (4)	80. (3)	116. (3)	152. (1)
9. (3)	45. (2)	81. (2)	117. (1)	153. (1)
10. (1)	46. (1)	82. (3)	118. (3)	154. (2)
11. (2)	47. (2)	83. (2)	119. (2)	155. (2)
12. (3)	48. (2)	84. (1)	120. (2)	156. (3)
13. (2)	49. (4)	85. (1)	121. (4)	157. (3)
14. (4)	50. (2)	86. (1)	122. (1)	158. (4)
15. (3)	51. (3)	87. (1)	123. (2)	159. (3)
16. (1)	52. (3)	88. (4)	124. (3)	160. (2)
17. (3)	53. (3)	89. (1)	125. (2)	161. (4)
18. (2)	54. (1)	90. (1)	126. (2)	162. (1)
19. (4)	55. (1)	91. (4)	127. (2)	163. (1)
20. (4)	56. (1)	92. (1)	128. (3)	164. (4)
21. (1)	57. (1)	93. (3)	129. (1)	165. (3)
22. (3)	58. (3)	94. (3)	130. (1)	166. (2)
23. (2)	59. (2)	95. (2)	131. (1)	167. (3)
24. (2)	60. (3)	96. (2)	132. (3)	168. (2)
25. (2)	61. (1)	97. (4)	133. (2)	169. (4)
26. (1)	62. (3)	98. (4)	134. (4)	170. (4)
27. (2)	63. (1)	99. (3)	135. (1)	171. (4)
28. (2)	64. (1)	100. (4)	136. (1)	172. (3)
29. (2)	65. (2)	101. (2)	137. (1)	173. (3)
30. (1)	66. (1)	102. (1)	138. (2)	174. (4)
31. (2)	67. (2)	103. (4)	139. (3)	175. (1)
32. (2)	68. (4)	104. (4)	140. (1)	176. (1)
33. (4)	69. (2)	105. (1)	141. (4)	177. (4)
34. (1)	70. (3)	106. (1)	142. (3)	178. (1)
35. (2)	71. (2)	107. (2)	143. (4)	179. (4)
36. (1)	72. (3)	108. (3)	144. (2)	180. (2)

Hints and Solutions

PHYSICS

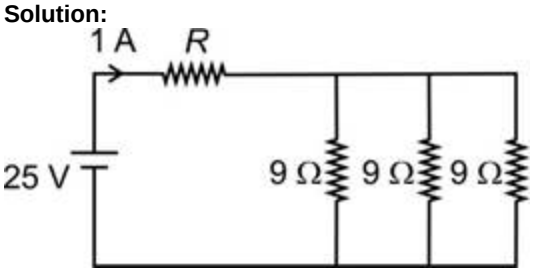
(1) **Answer :** (2)
Solution:
As we know that resistance of metals increases with increase in temperature. At same voltage $i_A < i_B$, so $R_A > R_B$, hence $T_A > T_B$.

(2) **Answer :** (2)
Solution:



then, $R_{eq} = \frac{12\times4}{12+4} \Rightarrow \frac{48}{16}$ = 3 Ω

(3) **Answer :** (1)
Hint:
Kirchhoff's laws
Solution:



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R_{net} =
$\frac{V}{i} = \frac{25}{i} = 25\ \Omega$

also,
$R_{net} = \frac{9}{3} + R = 25 \Rightarrow R = 22\ \Omega$

(4) **Answer :** (2)
Solution:
B and *D* are terminals of battery hence V_{BD} = 10 V

(5) **Answer :** (4)
Solution:
$p = \frac{v^2}{R}$, as *v* is same, $p \propto \frac{1}{R}$

(6) **Answer :** (3)
Solution:
Current density
$J = nev_d = \frac{I}{A}$
$v \propto \frac{I}{A}$ $\quad\quad\quad\quad$ $\therefore\ v = v$

$v' \propto \frac{(2I)}{2A} \Rightarrow v' = v$

(7) **Answer :** (1)
Solution:

$\because V = IR = neAV\left[\frac{p\ell}{A}\right] = neV\ell_p$

[V → drift speed]

$\therefore \rho = \frac{V}{neVl} = \frac{4}{6\times10^{28}\times1.6\times10^{-19}\times1.5\times10^{-3}\times0.2}$

ρ = 1.4 × 10^{-6} Ωm.

(8) **Answer :** (1)
Solution:
Current will be constant along the conductor.

(9) **Answer :** (3)
Solution:
Q = Area under *i* – *t* curve

$\boxed{Q = \frac{1}{2}i_0T}$

(10) **Answer :** (1)
Solution:

$R = \frac{\rho l}{a} = \frac{l}{\sigma a} = \frac{60\times10^{-2}}{6\times10^{7}\times10^{-4}} = 10^{-4}\Omega$

(11) **Answer :** (2)
Solution:

$R = \frac{ml}{ne^2A\tau}$

$R \propto \frac{1}{n}$

when *L* = constant, *A* = constant, τ = constant $\frac{R_A}{R_B} = \frac{n_B}{n_A} = \frac{3}{1}$

(12) **Answer :** (3)
Solution:
For (a) when length is 4 times, new length becomes 4*l* and area becomes $\frac{A}{4}$.

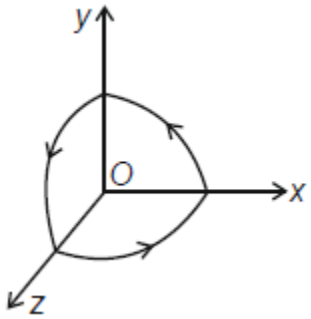
$R' = \rho\left(\frac{4l}{A/4}\right) = 16R$

(b) if wire is stretched to double its length $R \propto l^2$, $R' \propto (2l)^2 \therefore R' = 4R$.

(c) If radius of wire is halved $R \propto \frac{1}{r^2} \therefore R' = 4R$.

(d) If wire is bent to form L-shaped wire its resistance will remain R only.

(13) **Answer :** (2)
Solution:



$B_0 = \sqrt{3}\,B_{arc}$

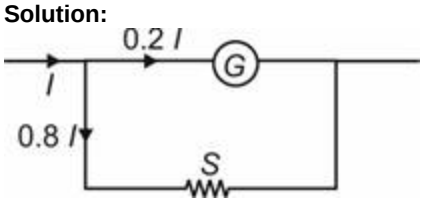
= $\sqrt{3}\left(\frac{\mu_0 i}{4(2R)}\right)$

$B_0 = \frac{\sqrt{3}\mu_0 i}{R}$

(14) **Answer :** (4)
Solution:

If $\vec{B} \perp \vec{v}$, path is circular.

(15) **Answer :** (3)
Solution:



0.2 I × 80 = 0.8 $I.S$
S = 20 Ω

(16) **Answer :** (1)
Solution:
Lorentz force
$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$
$\vec{F} = 0$, if particle was unaccelerated.
There should be no presence of electric field so that particle goes unaccelerated. Magnetic field may or may not accelerate particle.

(17) **Answer :** (3)
Solution:

According to Biot-Savart's law $d\vec{B} = \frac{\mu_0}{4\pi}\frac{id\vec{l} \times \vec{r}}{r^3}$ which is applicable for infinitesimal current element. It is analogous to Coulomb's law, where electric field is produced by scalar source q.

(18) **Answer :** (2)
Solution:
$M = \frac{QL}{2m}$
$\frac{M}{L} = \frac{Q}{2m}$

(19) **Answer :** (4)
Solution:
$U = -MB\cos\theta$
$U_i = -MB\cos\theta$
$= -MB$
$U_f = -MB\cos\theta$
$W = U_f - U_i$
$= -MB\cos\theta + MB$
$= MB(1 - \cos\theta)$

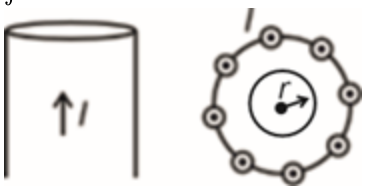
(20) **Answer :** (4)
Solution:
$\vec{F}_m = q\left(\vec{V} \times \vec{B}\right)$
$F \perp V$ & $F \perp B$

(21) **Answer :** (1)
Solution:
$B_1 = \frac{\mu_0 IR^2}{2\left(R^2+x^2\right)^{\frac{3}{2}}}$
$B_2 = \frac{\mu_0 l'}{2R}$
$\frac{B_1}{B_2} = \frac{R^3}{\left(R^2+x^2\right)^{\frac{3}{2}}}$
$B_2 = 250\mu T$

(22) **Answer :** (3)
Hint:
Use Ampere's circuital law.
Solution:

$\oint \vec{B} . \vec{dl} = \mu_0 i_{enc}$

$\oint \vec{B} . \vec{dl} = 0$

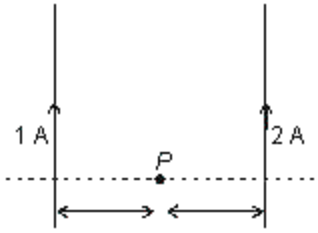


$B2\pi r = 0$
(where r < radius of pipe)
$B = 0$

(23) **Answer :** (2)
Solution:
$qE = qVB$
$v = \frac{E}{B}$

(24) **Answer :** (2)
Solution:
Energy of loop (dipole) in magnetic field is given by
$U = -MB\cos\theta$
For stable equilibrium U = min
For unstable equilibrium U = max

(25) **Answer :** (2)
Solution:

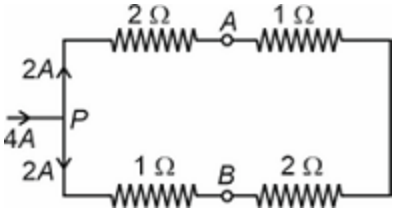


$B_P = B_{2A} - B_{1A} = B$ (given) ...(i)
and $B_{2A} = 2B_{1A}$ ...(ii)
From eq. (i) and (ii)
$B_{1A} = B$
So, after switching off
$\Rightarrow B_P = B_{2A} = 2B$

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(26) **Answer :** (1)
Solution:
Susceptibility of ferromagnetic material is large and positive.

(27) **Answer :** (2)
Solution:



Apply K.V.L
$V_P - 2 \times 2 = V_A$
$V_P - 2 \times 1 = V_B$
$V_A - V_B = V_P - 4 - V_P + 2$
$V_A - V_B = -2$ V

(28) **Answer :** (2)
Solution:



$\Rightarrow$ $\overline{M_0 \text{ (Let)}}$

$\because\ M = \frac{2M_0}{\pi}$

So $M_0 = \frac{\pi M}{2}$

(29) **Answer :** (2)
Solution:

$M_{net} = \sqrt{M_1^2 + M_2^2}$

$M_1 = M_2 = \frac{M}{2}$

$M_{net} = \frac{M}{\sqrt{2}}$

(30) **Answer :** (1)
Solution:
Area of hysteresis loop is energy loss per unit volume.

(31) **Answer :** (2)
Solution:

$\chi_m = \frac{I}{H}$

$I = \chi_m H = 3 \times 10^{-4} \times 4 \times 10^{-4}$

$= 1.2 \times 10^{-7}$ A m^{-1}.

(32) **Answer :** (2)
Solution:

$B_{\text{axis}} = \frac{\mu_0}{4\pi}\frac{2M}{r^3}$, $B_{\text{eq.}} = \frac{\mu_0}{4\pi}\frac{M}{r^3}$

$\frac{B_{\text{axis}}}{B_{\text{eq.}}} = \frac{2}{1}$

(33) **Answer :** (4)
Solution:
$I = \chi H$
For paramagnetic material, χ is positive, therefore magnetisation is in the direction of applied field.

(34) **Answer :** (1)
Solution:

$\text{Intensity of magnetisation} = \frac{\text{Magnetic moment}}{\text{Volume}}$

$I = \frac{6}{30 \times 10^{-6}} = \frac{10^6}{5} = 2 \times 10^5 \text{A/m}$

(35) **Answer :** (2)
Solution:

$I = \chi H \Rightarrow \chi = \frac{I}{H} \Rightarrow$ Ratio $\rightarrow$ Susceptibility

(36) **Answer :** (1)
Solution:

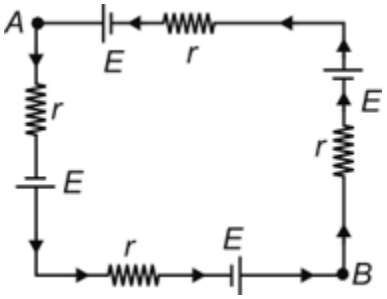
$i = \frac{q_{net}}{t}$

$= (n_1 + n_2)e$

Current flow in the direction of flow of positive charge and in opposite direction of flow of negative charge.

(37) **Answer :** (1)
Solution:

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$i = \frac{E}{R}$

$V_A - E + ir - E + ir = V_B$

$V_A - E + E - E + E = V_B$

$V_A = V_B$

(38) **Answer :** (2)
Solution:
At temperature much lower than 0°C, graph deviates considerably from a straight line.

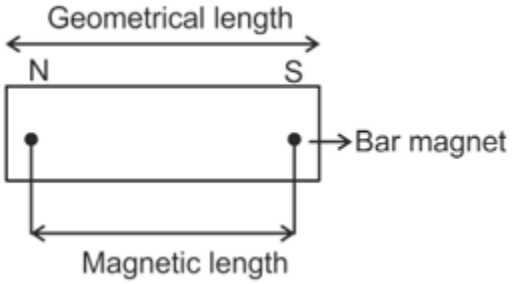
(39) **Answer :** (4)
Solution:
Semi-conductor have negative temperature coefficient of resistance.

(40) **Answer :** (1)
Solution:
$R \leq d$
$\frac{mv}{qB} \leq d$
$B_{\min} = \frac{v}{sd}$

(41) **Answer :** (3)
Solution:
To make the magnetic field radial in a moving coil galvanometer, pole magnet are cylindrical cut.

(42) **Answer :** (3)
Solution:
$n = 1000 \quad l = \frac{1}{4\pi}A$
$B = \mu_0 ni = 4\pi \times 10^{-7}(1000) \times \frac{1}{4\pi}$
$B = 10^{-4}$ (T)

(43) **Answer :** (2)
Solution:



Magnetic length = 0.84 × geometrical length

(44) **Answer :** (4)
Solution:
$B_m = \frac{eh}{4\pi m}$

(45) **Answer :** (2)
Hint:
R_2, R_3 are in series and then R_1 in parallel and $I = V/R_{eq} \quad \Rightarrow \quad i = \frac{4.8}{\frac{12}{5}} = 2\ \text{A}$

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CHEMISTRY

(46) **Answer :** (1)
Solution:
For zero order reaction
$r = k[A]^0$
Hence rate is independent to concentration

(47) **Answer :** (2)
Solution:
⇒ For first order reaction,
⇒ $K = \frac{0.693}{t_1}$
⇒ $t_{\frac{1}{2}} = \frac{0.693}{K} = \frac{0.693}{6.93\times10^{-2}} = 10 \text{ min}$

(48) **Answer :** (2)
Solution:
$r = k[X]^a[Y]^b$
$1.2 \times 10^{-3} = k[0.1]^a[0.2]^b$
$3 \times 10^{-2} = k[0.1]^a[0.1]^b$
$2.4 \times 10^{-3} = k[0.2]^a[0.2]^b$
(I) ÷(II)
$2^b = 4$
b = 2
(I) ÷(III)
$\left(\frac{1}{2}\right)^a = \frac{1}{2}$
a = 1
$r = k[X][Y]^2$

(49) **Answer :** (4)
Solution:
Rate of reaction
$-\frac{\text{d[A]}}{\text{dt}} = -\frac{1}{2}\frac{\text{d[B]}}{\text{dt}} = \frac{1}{4}\frac{\text{d[C]}}{\text{dt}} = \frac{1}{3}\frac{\text{d[D]}}{\text{dt}}$

(50) **Answer :** (2)
Hint:
For the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$
Rate of reaction $= -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$
Solution:
$-\frac{1}{3}\frac{\text{d[H}_2\text{]}}{\text{dt}} = \frac{1}{2}\frac{\text{d[NH}_3\text{]}}{\text{dt}}$
$\frac{1}{3}\times\frac{2}{1} = \frac{1}{2}\frac{\text{d[NH}_3\text{]}}{\text{dt}}; \quad \left[\frac{-\text{d[H}_2\text{]}}{\text{dt}} = \frac{4\text{g}}{\text{h}} = \frac{2 \text{ mol}}{\text{h}}\right]$
$\frac{\text{d[NH}_3\text{]}}{\text{dt}} = \frac{4}{3}\frac{\text{mol}}{\text{h}}$
Rate of change of NH_3 in g/h
$= \frac{4}{3}\times17 = 22.7 \text{ g/h}$

(51) **Answer :** (3)
Solution:
Rate = k[C]
$\left[\text{C}\right] = \frac{7\times10^{-4}}{2\times10^{-3}} = 0.35 \text{ M}$

(52) **Answer :** (3)
Solution:
$\frac{r_2}{r_1} = 2^{\frac{30}{10}} = 8$

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$r_2 = 8r_1$

(53) **Answer :** (3)
Solution:
For 1st order reaction

$t_{1/2}$	$\frac{\ln 2}{k}$
$t_{75\%}$	$\frac{\ln 4}{k}$
$t_{99\%}$	$\frac{2\ \ln 10}{k}$
$t_{90\%}$	$\frac{\ln 10}{k}$

(54) **Answer :** (1)
Hint:
Unit of k = $(mol/lit)^{1-n}\ s^{-1}$
Solution:
For zero order reaction, unit of k = mol $L^{-1}\ s^{-1}$ Unit of rate = mol $L^{-1}\ s^{-1}$

(55) **Answer :** (1)
Solution:
Molecularity of a reaction cannot be zero or a non-integer.

(56) **Answer :** (1)
Solution:
Catalyst does not catalyse non-spontaneous reactions.

(57) **Answer :** (1)
Solution:
$\triangle_r H = (E_a)_f - (E_a)_b$
$= 40 - 60$
$= -20\ kJ\ mol^{-1}$

(58) **Answer :** (3)
Hint:
Arrhenius equation :
$\ln k = \ln A - \frac{E_a}{RT}$
Solution:
Compare equation, $\ln k = 2 - \frac{200}{T}$
with Arrhenius equation
$\ln A = 2$ and $\frac{E_a}{R} = 200$
$\Rightarrow A = e^2$ and $E_a = 200\ R$

(59) **Answer :** (2)
Solution:
Arrhenius equation
$k = Ae^{-\frac{Ea}{RT}}$
$\ln k = \ln A - \frac{Ea}{R}\left(\frac{1}{T}\right)$
This graph will be straight line with negative slope.

(60) **Answer :** (3)
Solution:

$$\log \frac{4.5 \times 10^7}{1.5 \times 10^7} = \frac{E_a}{2.303 \times 8.314}\left[\frac{373-323}{373 \times 323}\right] \qquad \left\{\begin{array}{l} T_1 = 50 + 273 = 323 \\ T_2 = 100 + 273 = 373 \end{array}\right\}$$

$$\log\ 3 = \frac{E_a}{2.303 \times 8.314}\left[\frac{50}{373 \times 323}\right]$$

$$\frac{0.4770 \times 2.303 \times 8.314 \times 373 \times 323}{50} = E_a$$

$E_a = 22007\ J\ mol^{-1} \approx 2.2 \times 10^4$

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(61) **Answer :** (1)
Solution:
Oxidation state of Cr in
CrO_4^{2-} = $x - 2 \times 4 = -2$; $x = +6$
Oxidation state of Cr in $Cr_2O_7^{2-}$ = $2x - 14 = -2$;
$x = +6$

(62) **Answer :** (3)
Solution:

Metals	$E^{\circ}_{M^{2+}/M}$ **(V)**
Ti	-1.63
V	-1.18
Cr	-0.90
Cu	$+0.34$

(63) **Answer :** (1)
Solution:

Catalyst	**Process**
$TiCl_4 + Al(CH_3)_3$	Polymerisation of Ethylene
V_2O_5	Oxidation of SO_2 in the manufacture of H_2SO_4
Ni complex	Polymerisation of alkynes
Iron catalyst	Haber's process

(64) **Answer :** (1)
Solution:
The transition metal ions which contain d^0 and d^{10} configuration, are colourless.

(65) **Answer :** (2)
Solution:

Element	**Ionization enthalpy (kJ mol^{-1})**
Cr	653
Mn	717
Cu	745
Zn	906

(66) **Answer :** (1)
Solution:
$3\,MnO_4^{2-} + 4H^+ \rightarrow 2\,MnO_4^- + MnO_2 + 2H_2O$
Zinc has minimum value of enthalpy of atomisation (126 kJ mol^{-1}) in 3*d* series.

(67) **Answer :** (2)
Solution:
Higher is the effective nuclear charge, greater is the hydration enthalpy.

M^{2+}	Co^{2+}	Fe^{2+}	Cr^{2+}	V^{2+}
$\Delta_{hyd}H$ (kJ mol^{-1})	–2079	–1998	–1925	–1895

(68) **Answer :** (4)
Solution:
Interstitial compounds are chemically inert, very hard and retain metallic conductivity.

(69) **Answer :** (2)
Solution:
$E^{\circ}_{Cu^{2+}/Cu}$ is positive, hence copper will not liberate H_2 gas on reaction with dilute HCl.

(70) **Answer :** (3)
Solution:

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Correct order of density is
Ni > Fe > Cr > Ti

(71) **Answer :** (2)
Hint:
As the size of lanthanoid ion decreases, the basicity of their hydroxides also decreases.
Solution:
Correct order of size of lanthanoid ions is
$Pr^{3+} > Nd^{3+} > Eu^{3+} > Gd^{3+}$
Therefore order of basicity of lanthanoid hydroxide is
$Pr(OH)_3 > Nd(OH)_3 > Eu(OH)_3 > Gd(OH)_3$

(72) **Answer :** (3)
Solution:
$2\,\mathrm{Mn}\,O_4^- + 5SO_3^{2-} + 6H^+ \rightarrow 2\,\mathrm{Mn}^{2+} + 5SO_4^{2-} + 3H_2O$

$\because$ 2 moles of $\mathrm{MnO_4^-}$ = 5 moles of $\mathrm{SO_3^{2-}}$

$\therefore$ 2 mole $\mathrm{SO_3^{2-}}$ = $\frac{2}{5} \times 2 = \frac{4}{5}$ mole of $\mathrm{MnO_4^-}$

(73) **Answer :** (2)
Solution:
Gadolinium = $\left[\mathrm{Xe}\right] 4f^7 5d^1 6s^2$

(74) **Answer :** (1)
Hint:
Paramagnetic species contain unpaired electron.
Solution:

MnO_4^{2-}, $Mn^{6+} \Rightarrow [Ar]\ 4s^0 3d^1$

It contains one unpaired electron so behaves as paramagnetic species.

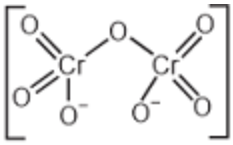
(75) **Answer :** (1)
Solution:
• Cerium (Ce) exhibits +2, +3 and +4 oxidation states while Eu and Yb exhibit +2 and +3 oxidation states.

(76) **Answer :** (1)
Solution:
Mischmetall contains 95% lanthanoid.

(77) **Answer :** (4)
Solution:

Ion	Radii (pm)
La^{3+}	106
Nd^{3+}	99
Ce^{3+}	103
Dy^{3+}	91

(78) **Answer :** (1)
Solution:



$\therefore$ 1 peroxide linkage present.

(79) **Answer :** (2)
Solution:

Element	Oxidation state(s)
Ac	+3
Th	+4

U	+3, +4, +5, +6
Np	+3, +4, +5, +6, +7

(80) **Answer :** (3)
Hint:
Curium is an actinoid.
Solution:
The correct electronic configuration of curium is $[Rn]5f^7 6d^1 7s^2$

(81) **Answer :** (2)
Solution:

$$A_2(g) \rightarrow B(g) + \tfrac{1}{2}C(g)$$

$t = 0$ $\quad 100\,mm \quad 0 \quad 0$ ($P_{initial}$ = 100, let x pressure dissociated)

$t = 5\ min \quad 100 - x \quad x \quad \frac{x}{2}$

At equilibrium $100 - x + x + \frac{x}{2} = 120$

$\therefore$ x = 40

$\therefore$ Rate of disappearance of $A_2 = \frac{40\ mm}{5\ mm}$

= 8 mmHg min^{-1}

(82) **Answer :** (3)
Solution:
For homogenous elementary reaction, order = number of reacting molecules.
A + B → C
order ⇒ 2
rate = k[A][B]
or, mol $L^{-1}s^{-1}$ = k mol^2L^{-2}
or, k = $\frac{mol\ L^{-1}s^{-1}}{mol^2\ L^{-2}}$ = $mol^{-1}Ls^{-1}$

(83) **Answer :** (2)
Solution:
r = K[RCl] → (r) doesn't depend on [NaOH]
r ∝ [RCl]
i.e., if [RCl]′ = $\left(\frac{RCl}{2}\right)$ then $r' = \frac{r}{2}$

(84) **Answer :** (1)
Solution:

1. Radioactive decomposition is first order reaction.
   So, $k = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{t} \log \frac{100}{100-x}$
   $\therefore \quad t_{1/2} = \frac{0.693}{k}$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{345} \log \frac{100}{100-80}$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{345} \log \frac{100}{20}$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{345} \log 5$

$\therefore$ [0.693 = 2.303 × log2]

$\frac{2.303 \times \log 2}{t_{1/2}} = \frac{2.303}{345} \log 5$

$t_{1/2} = 345 \times \frac{\log 2}{\log 5}$ = $345 \times \frac{\ln 2}{\ln 5}$

(85) **Answer :** (1)
Solution:
$[R]_0 - [R]_t = kt$
On completion of reaction, we can write a = kt $\quad [\because [R]_0 = a]$
or, $t = \frac{a}{k}$

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(86) **Answer :** (1)
Solution:
Gun metal contains (Cu, Sn, Zn)

(87) **Answer :** (1)
Solution:
$$K_2Cr_2O_7 + H_2SO_4 + 4H_2O_2 \rightarrow \underset{\text{Blue}}{2CrO_5} + K_2SO_4 + 5H_2O$$

(88) **Answer :** (4)
Solution:
Highest oxidation state of Fe is +6 and $Fe^{6+} \Rightarrow 3d^2$.

(89) **Answer :** (1)
Solution:
Au + Aqua regia → H [$AuCl_4$].

(90) **Answer :** (1)
Solution:
Transuranic elements are the elements, having atomic number greater than 92.
Atomic number of Np = 93
⇒ Transuranic elements begin with Np.

BOTANY

(91) **Answer :** (4)
Solution:
Chromosomal complement found in patients afflicted with Klinefelter's syndrome is 44 + XXY

(92) **Answer :** (1)
Hint:
In any cross, recombination frequency (RF) is equal to distance between genes.
Solution:
If RF is 15% it means the distance between gene is 15 cM.

(93) **Answer :** (3)
Solution:
It shows
(1)Pleiotropy.
(2) Incomplete dominance (Bb form intermediate size starch grains).
(3) Complete dominance (Bb, seed shape is round).

(94) **Answer :** (3)
Hint:
i allele is recessive to I^A and I^B.
Solution:
$I^AI^B \times I^Ai$

$$\text{Genotype}: \quad I^AI^A \quad I^Ai \quad I^BI^A \quad I^Bi$$
$$\text{Blood groups}: \quad A \quad , \quad A \quad , \quad AB \quad , \quad B$$

(95) **Answer :** (2)
Solution:
- Incomplete dominance – Phenotype did not resemble either of the two parents and was in between the two
- Pleiotropy – Phenylketonuria
- Codominance – F_1 resembles both parents
- Autosomal dominant trait – Myotonic dystrophy

(96) **Answer :** (2)
Solution:
In honeybee, haploid individuals are male and diploid ones are female. Honeybee shows haplodiploid sex determination.

(97) **Answer :** (4)
Solution:
The term used for the units of inheritance by Mendel was factors.

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...[1 Mark]

(98) Answer : (4)
Solution:
Mendel selected 14 true breeding pea varieties. His experiment had a large sampling size, which gave great credibility to the data that he collected.

(99) Answer : (3)
Solution:
α-Thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent and it is observed due to mutation or deletion of one or more of the four genes.

(100) Answer : (4)
Solution:
Constricted pod is a recessive trait of pea.

(101) Answer : (2)
Solution:
Punnett square is a graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross.

(102) Answer : (1)
Solution:
Point mutation occurs due to change in a single base pair of DNA.

(103) Answer : (4)
Solution:
Test cross is the cross of hybrid (unknown genotype) having dominant phenotype with homozygous recessive parent.

(104) Answer : (4)
Solution:
Drosophila exhibits clear differentiation of the sexes-the male and female flies are easily distinguishable.

(105) Answer : (1)
Solution:
Slightly different forms of the same gene are called alleles.

(106) Answer : (1)
Solution:
Since, genotype of individual is AaBBCc
Hence, number of gamete = 2^n (n = no. of heterozygous pairs)
$= 2^2 = 4$

(107) Answer : (2)
Hint:
Phenylketonuria is a Mendelian disorder.
Solution:
Turner's syndrome is caused due to the absence of one X-chromosome. Down's syndrome is characterised by broad palm with characteristic palm crease. Gynaecomastia is a symptom of Klinefelter's syndrome and phenylketonuria is also known as inborn error of metabolism.

(108) Answer : (3)
Solution:
Red eye colour in *Drosophila* is an X-linked dominant trait.

(109) Answer : (3)
Solution:
Law of segregation is also called law of purity of gametes. According to this law, factors of a character get separated from each other during gamete formation.

(110) Answer : (2)
Solution:
Phenotype is an observable morphological appearance of an organism.

(111) Answer : (3)
Solution:

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$XY \quad X \quad X^h \; X$ (parents)

XX^h, XX, X^hY, XY (off springs)

haemophilic 25%

(112) Answer : (1)

Hint:

Chromosomal aberration includes change in chromosome structure.

Solution:

Transition, frameshift mutation and deletion of two bases are included under gene mutation. Deletion of a part of a chromosome is included in aberration.

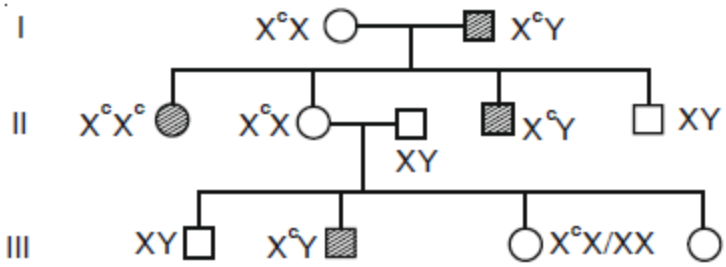
(113) Answer : (2)

Hint:

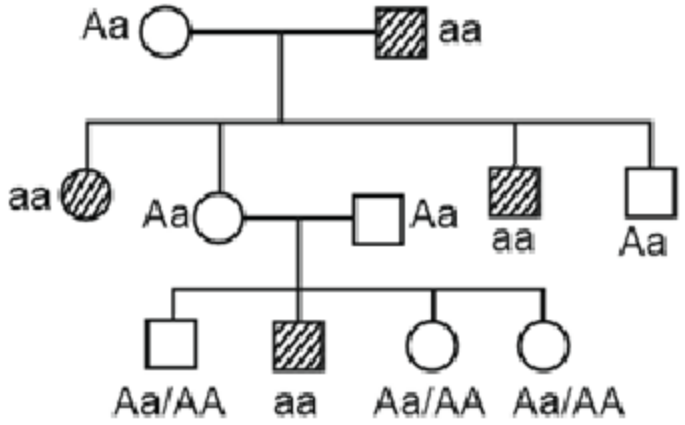
The given pedigree is applicable for an autosomal as well as for an X-linked recessive, but it does not show Y-linked inheritance as the female is also affected in second generation.

Solution:

If pedigree is showing X-linked inheritance–



If pedigree is showing an autosomal disorder–



So the given pedigree is applicable for an autosomal as well as X-linked recessive disorders.

(114) Answer : (3)

Hint:

Phenylalanine hydroxylase converts the amino acid phenylalanine into tyrosine.

Solution:

Phenylketonuria affected individual lacks a liver enzyme called phenylalanine hydroxylase which converts phenylalanine into tyrosine.

(115) Answer : (4)

Solution:

Queen Victoria was a carrier of the disease haemophilia.

(116) Answer : (3)

Solution:

The law of independent assortment states that the inheritance of one trait is independent of the inheritance of another trait, provided the genes are located on different chromosomes. This law is a fundamental principle of genetics proposed by

Gregor Mendel, not Bateson. It is best explained using a dihybrid cross, not a monohybrid cross, and it is not universally applicable as it does not hold true for genes that are linked (located close together on the same chromosome).

(117) Answer : (1)
Solution:
Gametes contain haploid number of chromosomes and each chromosome has many alleles.

(118) Answer : (3)
Solution:

$$\underset{\text{(pink)}}{\text{Rr}} \times \underset{\text{(red)}}{\text{RR}} \rightarrow \underset{\text{(pink)}}{\text{Rr}} : \underset{\text{(red)}}{\text{RR}} \quad \text{Ratio } \boxed{1:1}$$

(119) Answer : (2)
Solution:
2n + 1 represents trisomy.

(120) Answer : (2)
Hint:
In 1900, de Vries, Correns and von Tschermak independently rediscovered Mendel's results on the inheritance of characters.
Solution:
Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome. T.H. Morgan coined the term linkage to describe this physical association of genes on a chromosome. Henking could trace a specific nuclear structure all through spermatogenesis in a few insects.

(121) Answer : (4)
Solution:
Phenotypes in dihybrid test cross.

YyRr × yyrr	↓ ↓ yr
YR	YyRr
Yr	Yyrr
yR	yyRr
yr	yyrr

} 4 genotypes and 4 phenotypes

(122) Answer : (1)
Solution:
Recessive allele is a modified allele. It may produce non-functional enzyme.

(123) Answer : (2)
Solution:
Variation refers to degree by which progeny differ from their parents.

(124) Answer : (3)
Solution:
Probability of three sons $= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

(125) Answer : (2)
Solution:
Since, probability of inflated pod = $\frac{3}{4}$; and probability of white coloured flower = $\frac{1}{4}$
$\Rightarrow \frac{3}{4} \times \frac{1}{4} \times 512 \Rightarrow 96$

(126) Answer : (2)
Solution:

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F_1 AaBb

A
- B → AB
- b → Ab

a
- B → aB
- b → ab

(127) Answer : (2)
Solution:
Individuals having homomorphic sex chromosomes produce one type of gametes.

(128) Answer : (3)
Solution:
Mutation → Sudden change in genetic Material
Recombination → Non-parental combination

(129) Answer : (1)
Solution:
 represents sex unspecified.

(130) Answer : (1)
Solution:
Genes located on chromosome → 12

(131) Answer : (1)
Solution:
Haemophilia is an X-linked recessive disorder.

(132) Answer : (3)
Solution:
Down's syndrome occurs due to trisomy of the 21st chromosome.

(133) Answer : (2)
Solution:
AabbCCddEe × AabbCcDdee
↓
AabbCcDdee
$\frac{2}{4} \times \frac{2}{4} \times \frac{2}{4} = \frac{8}{64} = 12.5$

(134) Answer : (4)
Solution:
Plant Height and Pod shape – Location of traits is on chromosome number 4

(135) Answer : (1)
Solution:
Mendel experimented on pea for 7 years (1856-1863).

ZOOLOGY

(136) Answer : (1)
Solution:
Saheli, the new oral contraceptive for the females, contains a non-steroidal preparation named ‘Centchroman’ which is a selective estrogen receptor modulator. It is a ‘once a week’ pill with very few side effects and high contraceptive value.

(137) Answer : (1)
Solution:
Cu^{2+} interfere in the sperm movement, hence suppress the sperm motility and fertilising capacity of sperms.

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(138) Answer : (2)
Solution:
Statutory ban on amniocentesis has been imposed for sex-determination to legally check increasing menace of female foeticides. In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the foetal cells and dissolved substances. This procedure is used to test for the presence of certain genetic disorders such as down syndrome, haemophilia, sickle cell anemia, *etc*., to determine the survivability of the foetus.

(139) Answer : (3)
Solution:
Progestasert and LNG-20 are hormone releasing IUDs which make the uterus unsuitable for implantation and the cervix hostile to sperms.

(140) Answer : (1)
Solution:
'Family planning' programmes were initiated in 1951 in India and were periodically updated and assessed over the past decades.

(141) Answer : (4)
Solution:
Unsafe sex increases the chances of transmission of HIV infection significantly.

(142) Answer : (3)
Solution:
GIFT stands for gamete intra-fallopian transfer.

(143) Answer : (4)
Solution:
Tubectomy is a sterilisation method for females. Condom is a barrier contraceptive.

(144) Answer : (2)
Solution:
IUDs are intra uterine devices which are fitted into the uterus.

(145) Answer : (2)
Solution:
According to the World Health Organisation (WHO), reproductive health means a total well-being in all aspects of reproduction, *i.e*., physical, emotional, behavioural and social. Therefore, a society with people having physically and functionally normal reproductive organs and normal emotional and behavioural interactions among them in all sex-related aspects might be called reproductively healthy.

(146) Answer : (2)
Solution:
Norplant is a progestin only device containing levonorgestrel. Implanon contains 3-keto desogestrel(progestin).

(147) Answer : (3)
Solution:
Contraceptive methods may cause side effects such as nausea, abdominal pain, irregular menstrual bleeding, *etc*.

(148) Answer : (1)
Solution:
Condoms and vasectomy are contraceptive methods for males.

(149) Answer : (3)
Solution:
During lactation, plasma concentration of prolactin increases which acts as GnRH inhibitor preventing follicular development and ovulation, resulting into amenorrhoea named lactational amenorrhoea.

(150) Answer : (1)
Solution:
Progestasert and LNG-20 are hormone releasing IUDs.

(151) Answer : (4)
Hint:
It is one of the natural method of contraception.
Solution:
The withdrawal method is also known as coitus interruptus.

(152) Answer : (1)
Solution:

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Steroidal oral contraceptive pills have to be taken daily for a period of 21 days starting preferably within the first five days of menstrual cycle. After a gap of 7 days (during which menstruation occurs), it has to be repeated in the same pattern till the female desires to prevent conception.

(153) Answer : (1)
Solution:
Periodic abstinence (rhythm method) is a natural method of contraception in which the couples avoid or abstain from coitus from day 10 to 17 of the 28 days menstrual cycle when ovulation could be expected.

(154) Answer : (2)
Hint:
ZIFT is Zygote Intra Fallopian Transfer.
Solution:
In GIFT *i.e*., Gamete Intra Fallopian Transfer, fertilization of sperm and egg occurs inside the female body, *i.e*., *in vivo*, while in ZIFT, the fertilization of sperm and egg occurs in a petridish in a laboratory *i.e*., *in vitro*.

(155) Answer : (2)
Solution:
Transfer of sperms into uterus is called intra uterine insemination.

(156) Answer : (3)
Solution:
Gonorrhoea, AIDS and chlamydiasis are sexually transmitted diseases.

(157) Answer : (3)
Solution:
Ideal contraceptives should be effective against pregnancy and must be reversible.

(158) Answer : (4)
Hint:
Barrier methods prevent contact of sperms with ovum.
Solution:
Diaphragm, vault and condom are barrier methods of contraception but combined pills are oral contraceptive pills (OCPs) which prevent pregnancy by inhibiting ovulation.

(159) Answer : (3)
Solution:
Lippes loop is an IUD.

(160) Answer : (2)
Solution:
Test tube baby programme involves collection of ova from the wife/donor (female) and sperms from the husband/donor (male) and are induced to form zygote under simulated conditions in the laboratory.

(161) Answer : (4)
Solution:
According to The Medical Termination of Pregnancy (Amendment) Act, 2017, a pregnancy may be terminated on certain considered grounds within the first 12 weeks of pregnancy on the opinion of one registered medical practitioner. If the pregnancy has lasted more than 12 weeks, but fewer than 24 weeks, two registered medical practitioners must be of the opinion, formed in good faith, that the required grounds exist.

(162) Answer : (1)
Solution:
MMR stands for maternal mortality rate.

(163) Answer : (1)
Hint:
Technique used *in-vivo* fertilization.
Solution:
Fertilization occurs in fallopian tube when both ova and sperms are placed in oviduct in GIFT.

(164) Answer : (4)
Hint:
Progesterone gives negative feedback to hypothalamus.
Solution:
Higher concentration of estrogen and progesterone during luteal phase gives negative feedback to hypothalamus, thereby inhibiting the release of GnRH from hypothalamus and LH, FSH from pituitary.

(165) Answer : (3)
Solution:

Hepatitis-B, genital herpes and HIV infections are viral diseases which are not curable, even if detected early and treated properly.

(166) Answer : (2)
Hint:
Method of contraception which is poorly reversible in females.
Solution:
In tubectomy, a part of fallopian tubes are cut and tied. As a result of this, transport of ova is blocked.

(167) Answer : (3)
Solution:
Worldwide nearly 20% of the total conceived pregnancies get aborted *via* MTPs.

(168) Answer : (2)
Solution:
After 6 months of parturition, prolactin level decreases and FSH and LH is released which causes formation of follicles and eventually ovulation takes place.

(169) Answer : (4)
Solution:
Improved health facilities and better living conditions promoted an explosive growth of population. Such a growth necessitated intense propagation of contraceptive methods.

(170) Answer : (4)
Solution:
IVF (*in-vitro* fertilization) followed by embryo transfer in her uterus will help the women with blocked fallopian tubes, as it completely bypasses the tubes by fertilizing egg outside the fallopian body and implanting the embryo directly into the uterus.

(171) Answer : (4)
Solution:
Lactic acid, boric acid and citric acid are active ingredients used in chemical methods of contraception.

(172) Answer : (3)
Solution:
Nirodh is a popular brand of condom for male.

(173) Answer : (3)
Solution:
Diaphragm is placed inside the vagina.

(174) Answer : (4)
Solution:
Depo-Provera is an injectable contraceptive.

(175) Answer : (1)
Solution:
Norplant is progestin only device having six small silicone capsules each having levonorgestrel.

(176) Answer : (1)
Solution:
Spermicidal creams, jellies and foams increase the contraceptive efficiency of barrier contraceptive methods.

(177) Answer : (4)
Solution:
The drugs used for emergency contraception are morning after pills, IUDs, *etc*.

(178) Answer : (1)
Solution:
Contraceptive devices should not interfere with arousal of sexual drive or desire. Castration refers to removal of testes in males.

(179) Answer : (4)
Solution:
As no medicines or devices are used in natural method of contraception, their side effects are almost nil. Chances of failure, though, of these methods are also high.

(180) Answer : (2)
Solution:
– Chlamydiasis is a bacterial STI.
– AIDS is caused by Human Immunodeficiency Virus (HIV).
– Genital herpes is caused by Herpes Simplex Virus II

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